

Use-Case Flow Specification

Use Case: Generate Weekly Performance Optimization Digest

Use Case ID

UC-04

Primary Actors

- Database Administrator
- Backend Lead

Preconditions

1. Slow query logs have been successfully ingested into the system.
2. Relevant query performance data has been analysed.
3. The system has identified performance issues and optimization recommendations.
4. The user is authorized to access the performance optimization digest.

Postconditions

1. A weekly performance optimization digest is generated successfully.
2. The digest contains identified slow queries, performance issues, and optimization recommendations.
3. The Database Administrator and Backend Lead can view the generated digest.

Main Success Scenario

1. The Database Administrator or Backend Lead accesses the Database Query Performance Profiler.

2. The user selects the option to generate the Weekly Performance Optimization Digest.
3. The system retrieves the slow query logs collected during the selected weekly reporting period.
4. The system analyses the query performance data and identifies significant performance issues.
5. The system retrieves the execution-plan analysis and index recommendations for the identified queries.
6. The system compiles the slow queries, detected performance issues, and recommended optimization actions.
7. The system generates the Weekly Performance Optimization Digest.
8. The system displays the generated digest to the authorized user.
9. The user reviews the query performance issues and recommended optimization actions.
10. The use case ends successfully.

Alternate Flow: No Performance Issues Found

4a. No significant performance issues are identified during the selected reporting period.

- 4a1. The system detects that no queries exceed the configured performance thresholds.
- 4a2. The system generates a digest stating that no significant database performance issues were identified.
- 4a3. The system displays the digest to the authorized user.
- 4a4. The use case ends successfully.